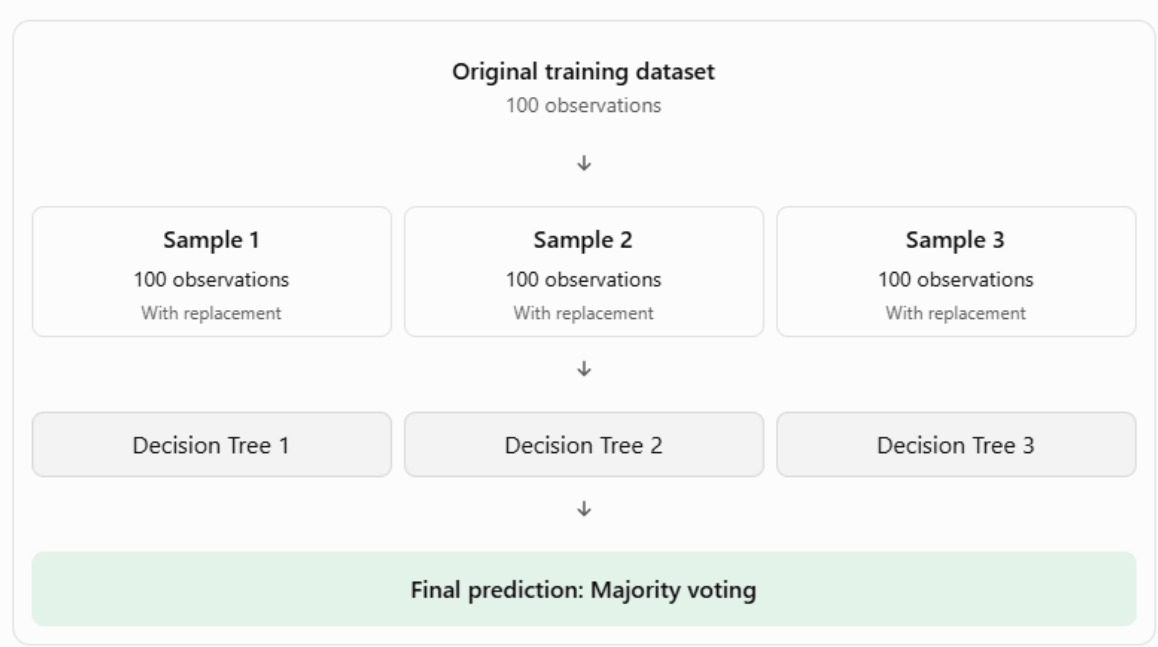
Combining Classifiers: Ensemble Learning

Combining Classifiers (Ensemble Learning)

- Ensemble learning is a machine learning technique that combines multiple classifiers to produce a more accurate and reliable prediction than a single classifier. The individual classifiers are called base learners or weak learners.
- The main idea is that combining the predictions of multiple models can reduce errors and improve generalization.
- The major ensemble learning techniques are:
  - Bagging (Bootstrap Aggregating)
  - Boosting
  - Random Forest
  - AdaBoost (Adaptive Boosting)

- **Bagging (Bootstrap Aggregating)**
  - Bagging is an ensemble technique in which multiple models are trained independently on different bootstrap samples of the original training dataset.
  - Bootstrap samples are generated by randomly selecting observations with replacement. Suppose the original dataset contains 100 observations. We generate three bootstrap samples, each containing 100 observations selected with replacement.



  - Each model produces a prediction, and the final prediction is obtained by combining the individual predictions.

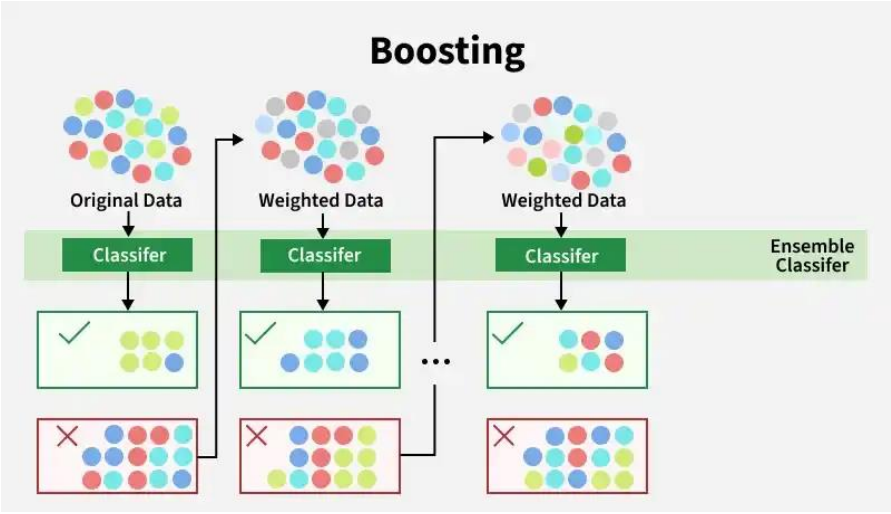
- Working procedure:

    1. Generate multiple bootstrap samples from the original dataset.
    2. Train a separate classifier on each sample.
    3. Obtain predictions from all classifiers.
    4. Combine predictions using majority voting for classification or averaging for regression.

- For classification, the final prediction is:

$$\hat{y} = \text{mode}\{\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_M\}$$

- **Boosting**
  - Boosting is an ensemble learning technique in which classifiers are trained sequentially (one after another).
  - Each new classifier attempts to correct the errors made by the previously trained classifiers.
  - Initially, all training samples are assigned equal importance. During subsequent iterations, greater emphasis is placed on incorrectly classified samples.
  - The final prediction is obtained by combining the predictions of all classifiers using weighted voting.

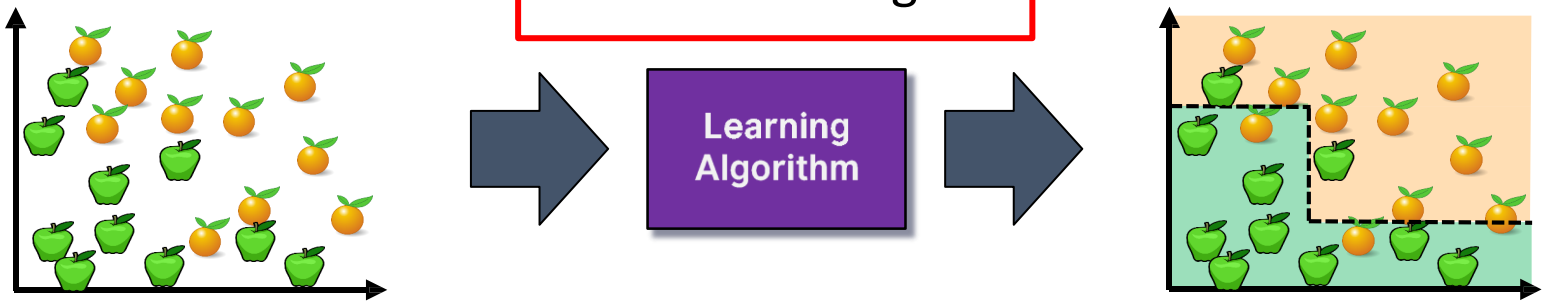


Working Procedure

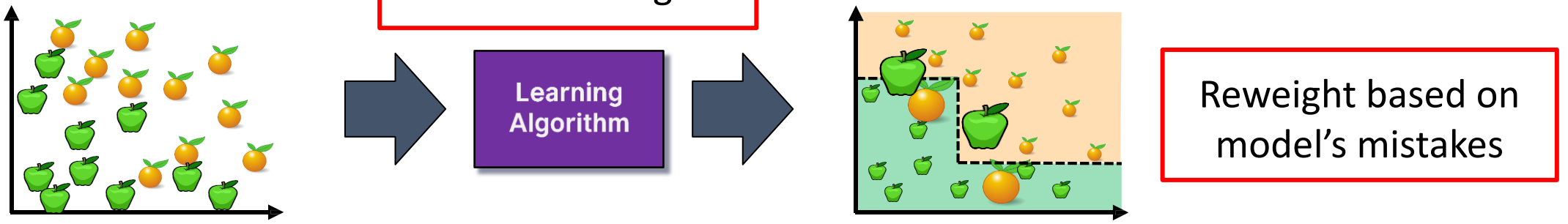
1. **Base Model:** A simple base model (usually a shallow Decision Tree, often called a "stump") is trained on the dataset.
2. **Calculate Residuals:** The algorithm evaluates the predictions against the actual target values to compute the errors (residuals).
3. **Train on Errors:** A subsequent model is trained *not* on the original dataset, but on the errors of the previous model.
4. **Weighted Combination:** Combine the predictions of all classifiers, often using a weighted vote or weighted sum, to produce the final prediction.

Boosting

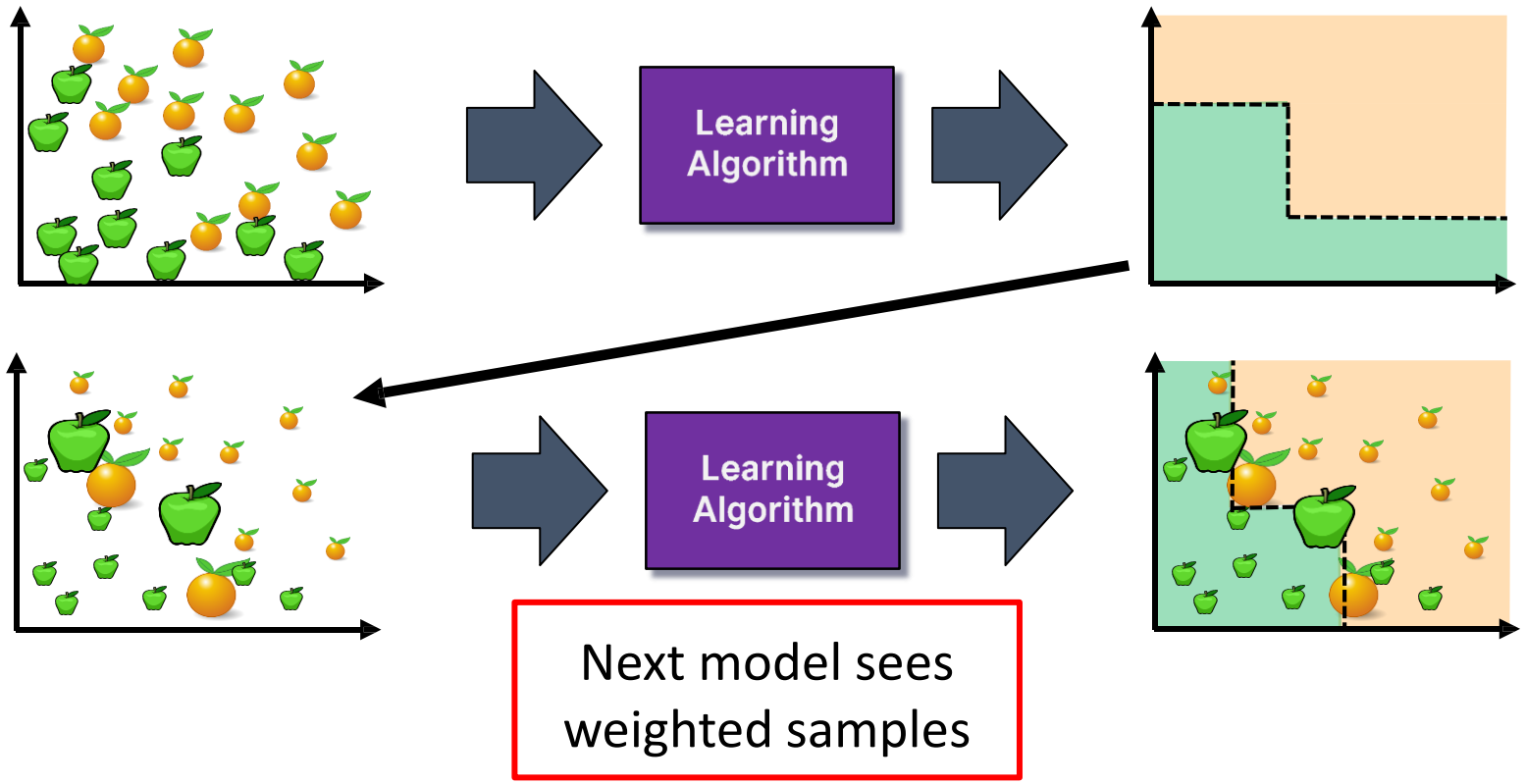
All samples have the same weight



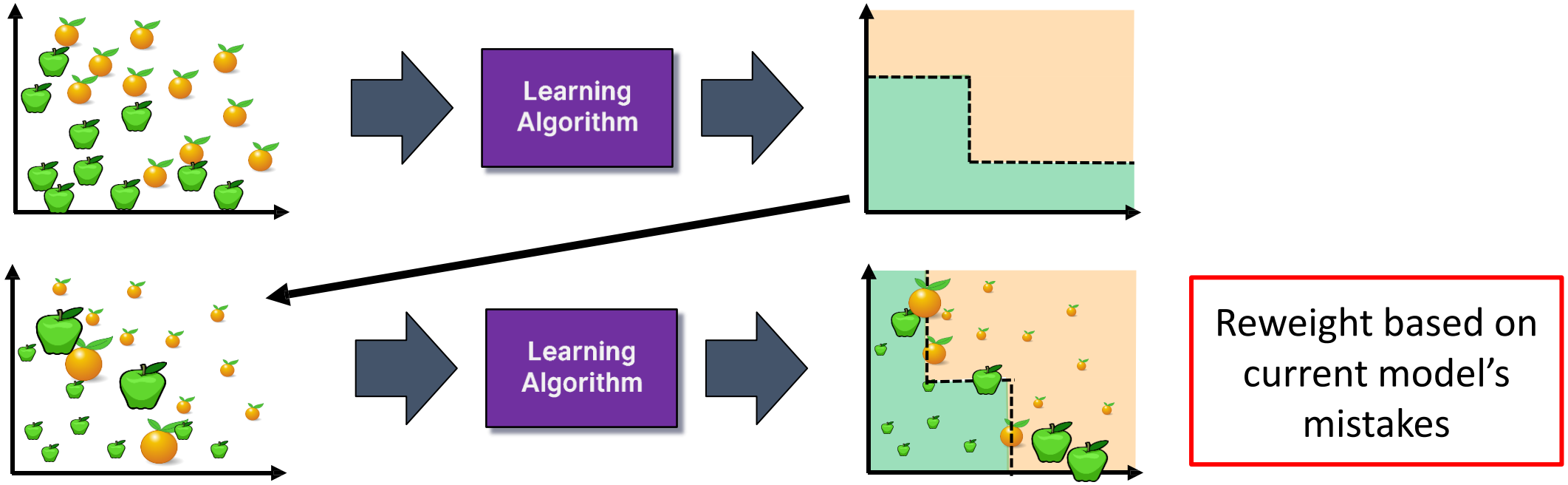
Boosting


All samples have the same weight
Learning Algorithm
Reweight based on model's mistakes

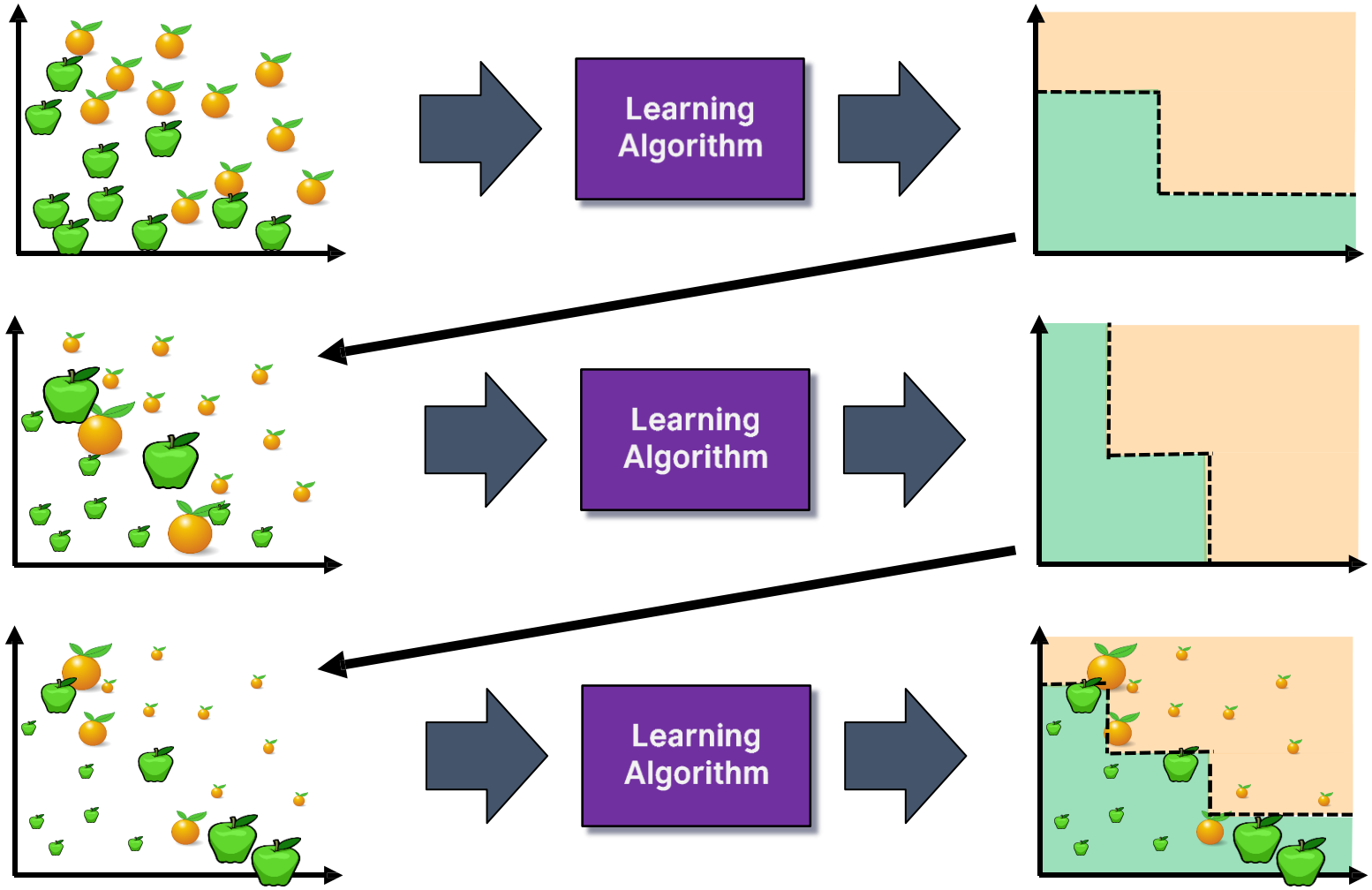
Boosting


Learning Algorithm
Learning Algorithm
Next model sees weighted samples

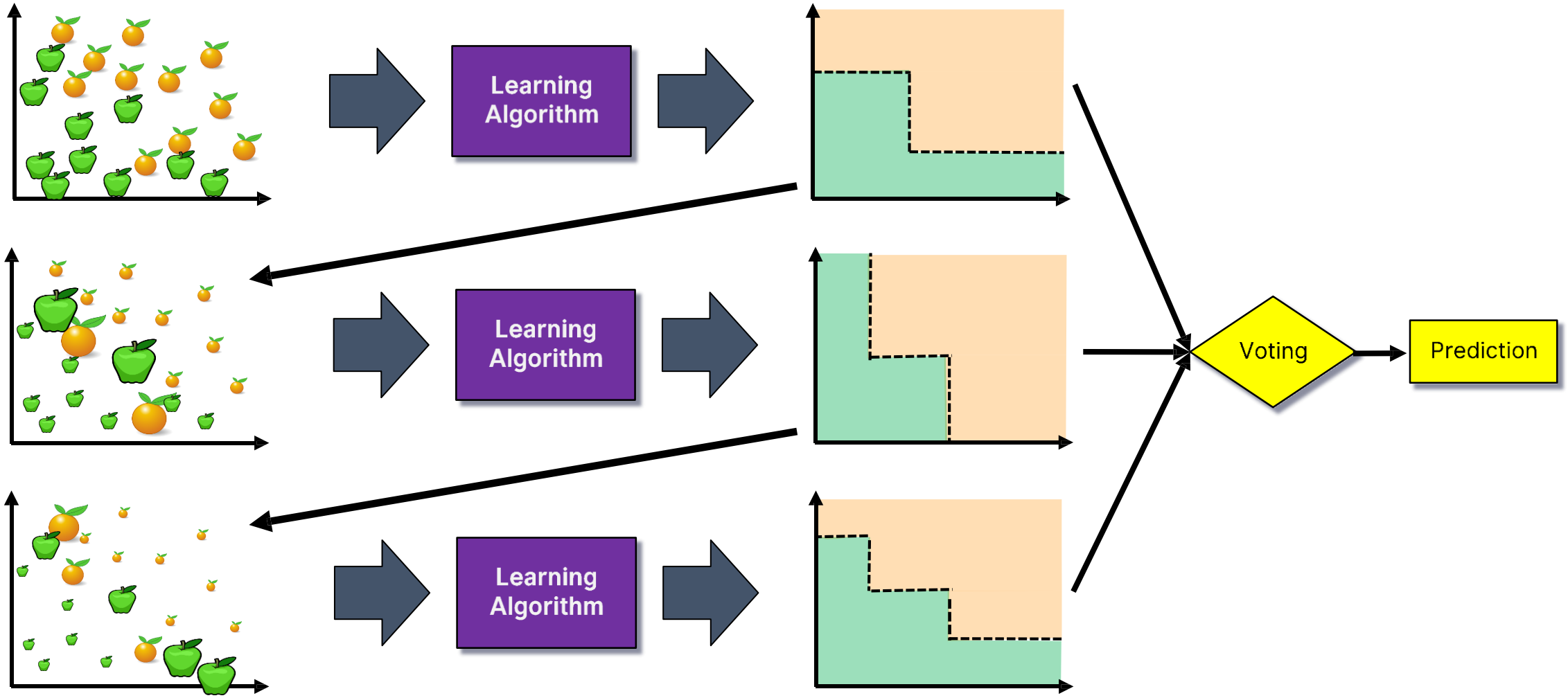
Boosting


Learning Algorithm
Learning Algorithm
Reweight based on current model's mistakes

Boosting


Learning Algorithm
Learning Algorithm
Learning Algorithm

Boosting


Learning Algorithm
Learning Algorithm
Learning Algorithm
Voting
Prediction

AdaBoost (Adaptive Boosting)

- AdaBoost is a powerful ensemble machine learning algorithm that combines multiple weak classifiers to build a highly accurate strong classifier.
- Developed by Yoav Freund and Robert Schapire in 1995, it is called "adaptive" because subsequent weak learners are tweaked in favor of those instances misclassified by previous classifiers.

Working:

AdaBoost builds its final model sequentially through a series of iterations:

1. **Initialize Sample Weights:**
   Every training sample starts with an equal weight: $W_i = \frac{1}{N}$, where N is the total number of data points.
2. **Train a Weak Learner:**
   A weak classifier (typically a **Decision Stump**—a decision tree with only one split and two leaves) is trained on the data using the current sample weights.
3. **Calculate Error (ε):**
   The algorithm calculates the total weighted error of the weak learner by adding up the weights of the misclassified samples.
4. **Calculate Classifier Weight (α):**
   AdaBoost assigns an "amount of say" (importance score) to the weak learner using its error rate.

$$\alpha = \frac{1}{2} \ln\left(\frac{1 - \text{Error}}{\text{Error}}\right)$$

Accurate classifiers get a high positive score; inaccurate ones get a penalty.

- **Advantages**
  - Can improve predictive accuracy by combining weak learners.
  - Often reduces bias and can improve generalization.
  - Focuses learning on difficult observations.
- **Limitations**
  - Training is sequential, so parallelization across boosting stages is limited.
  - Can be sensitive to noise and outliers, particularly AdaBoost.
  - Excessive boosting may lead to overfitting.
- Key difference between bagging and Boosting: Bagging trains independent models on different samples and combines their predictions, whereas boosting trains models sequentially, with each stage attempting to improve the ensemble's performance.

Popular Boosting Algorithms

Algorithm	Key Characteristics	Best Use Cases
AdaBoost (Adaptive Boosting)	Focuses heavily on instances misclassified by previous models by increasing their weights iteratively.	Binary classification, simpler datasets.
Gradient Boosting (GBM)	Builds trees sequentially by directly optimizing a differentiable loss function using gradient descent.	General tabular data, regression, and classification tasks.
XGBoost (Extreme Gradient Boosting)	Highly optimized version of GBM featuring parallelization, tree pruning, and built-in regularization to prevent overfitting.	Kaggle competitions, large-scale tabular datasets, fast performance.
LightGBM	Developed by Microsoft; uses gradient-based one-side sampling (GOSS) and exclusive feature bundling (EFB) for ultra-fast training on massive datasets.	High-dimensional data, massive datasets, speed-critical applications.
CatBoost	Developed by Yandex; handles categorical features automatically with minimal preprocessing and uses ordered boosting to reduce overfitting.	Datasets with many categorical variables.